LLM-Powered Supply Chain Risk and Fraud Detection

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GitHub Repository: https://github.com/ArdraJ0361/supply-chain-risk-detection

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Abstract

The role of pretrained text embeddings in fraud detection in supply chain system is studied in this research in comparison with conventional machine learning features. By means of DataCo SMART Supply Chain Dataset, it is considered structured XGBoost classifier, semi-supervised Isolation Forest, embedding-based XGBoost model with MiniLM embeddings, and the hybrid model, which combines structured and text embeddings. The data has been split according to Order Id to avoid overlap between train, validation, and test sets. The metrics of the models included Precision-Recall Area Under the Curve (PR-AUC), as well as ROC-AUC. Structured XGBoost model provided the result of 0.0765 PR-AUC and 0.8620 ROC-AUC, with recall of 66.47% and precision of 7.85% at a decision threshold of 0.70. Across three split seeds, the hybrid model's PR-AUC differed from structured XGBoost by at most 0.006 (+0.0059, +0.0013 and –0.00002), and a shuffled-embedding control showed no reliable benefit. As for the results, it became clear that all positive labels have referred to TRANSFER class. Thus, the results have been strongly affected by the particular label relations.

1. Introduction

The transaction data generated by supply chain management applications consists of many characteristics, such as product data, payment type, customer segment data, geographic data, shipping data, etc. Such transaction analysis may help identify potential frauds for investigation.

Traditional methods of machine learning involve analyzing the structured numerical or categorical features of the data to detect patterns. In contrast, language models learn to encode textual data into dense numerical embeddings. Combining both representations could help in detecting any extra semantic information. Nevertheless, usefulness of the text embeddings would depend upon the amount of textual data present in the dataset.

In this research, the DataCo SMART Supply Chain dataset is used to determine whether pre-trained text embeddings could help in the identification of suspicious transactions marked as frauds. In this study, comparison of structured machine learning, label-filtered (semi-supervised) Isolation Forest anomaly detection baseline, classification using text embeddings and hybrid models is performed.

The previous research provided sentence level embeddings to enable better use of semantic representations for similarity and other natural language processing (NLP) tasks (Reimers & Gurevych, 2019), while MiniLM uses self-attention distillation to build compactly pretrained Transformer

embeddings (Wang et al., 2020). In case of imbalanced classification, PR analysis may be more insightful than ROC analysis (Saito & Rehmsmeier, 2015).

Research Objectives

This study (1) compares structured XGBoost, Isolation Forest, MiniLM embedding-only, and hybrid models; (2) tests the effect of removing payment `Type`; and (3) evaluates split sensitivity and shuffled-embedding controls using metrics suited to an imbalanced target.

2. Dataset and Preprocessing

2.1 Dataset Description

The DataCo SMART Supply Chain Dataset was obtained from Kaggle. The dataset contains 180,519 records and 53 columns. The target variable was derived from the `Order Status` field: records labelled `SUSPECTED_FRAUD` were treated as positive examples, while all other records were treated as negative examples.

There were 4,062 positive records, representing approximately 2.25% of the dataset. This class imbalance makes accuracy alone an unsuitable measure of model performance.

Dataset source: DataCo SMART Supply Chain Dataset

2.2 Data Cleaning and Feature Selection

Initial inspection identified missing values in several columns. `Product Description` was entirely missing, while `Order Zipcode` was mostly missing. Unnecessary identifiers, personal information, and dummy contact fields were excluded from the predictors.

Potentially outcome-related or post-outcome features, including `Order Status`, `Delivery Status`, actual shipping duration, late-delivery risk, and shipping date, were excluded to reduce information leakage. `Order Id` was retained only for grouping records during dataset splitting.

Numerical features were imputed using the median. Missing categorical values were represented by a designated missing category, and categorical variables were one-hot encoded. The structured model used nine numerical and ten categorical predictors.

2.3 Data Splitting

The data were divided into training, validation, and test sets using a 60:20:20 split grouped by `Order Id` and stratified at the order level. This ensured that records belonging to the same order did not appear in multiple splits.

The split sizes were 108,360 training, 35,857 validation, and 36,302 test records (60:20:20), with no `Order Id` shared across splits. The test set contained 832 positive examples. Text embeddings were generated from eight product, category, customer-segment, market, geographic, and shipping fields.

3. Methodology

3.1 Structured XGBoost Classifier

XGBoost was used as the primary supervised classification model. It processed numerical features and one-hot-encoded categorical features, with missing numerical values imputed using the median. The classifier was configured with 300 estimators, a maximum tree depth of 5, a learning rate of 0.05, and a positive-class weighting strategy to address class imbalance.

3.2 Isolation Forest

Isolation Forest was evaluated as a label-filtered (semi-supervised) anomaly-detection baseline. It used nine numeric transaction and location features and 200 trees (`n_estimators=200`). Median imputation was fitted on the training data, and the detector was fitted only on training records labelled non-fraud. Anomaly scores were used to rank records, with the decision cutoff set to flag approximately the observed positive-class prevalence. Its performance was compared against the labelled validation data.

3.3 Text Embeddings

The pretrained `sentence-transformers/all-MiniLM-L6-v2` sentence encoder converted eight selected textual and categorical fields into 384-dimensional embeddings. For each record, fields were formatted as `Column: value` and joined with `|` (for example, `Product Name: ... | Category Name: ...`). MiniLM is a sentence encoder that produces embeddings, not a generative large language model. The embeddings were standardized using training-set statistics, with no PCA dimensionality reduction.

An embedding-only XGBoost classifier was trained to investigate whether these semantic representations could distinguish records labelled as suspected fraud without the structured feature set.

3.4 Hybrid Model and Feature Ablation

The hybrid model combined structured predictors with MiniLM embeddings. Its performance was compared with the structured-only and embedding-only approaches.

Feature-ablation experiments removed the payment transaction `Type` feature from the structured and hybrid models. This experiment investigated how strongly the observed results depended on payment type.

3.5 Evaluation Metrics and Threshold Selection

Average precision, reported as PR-AUC, was the primary evaluation metric because suspected-fraud records represented only a small proportion of the dataset. ROC-AUC, precision, recall, and F1-score were also reported.

The decision threshold was selected using the validation set by maximizing F1-score. The selected threshold of 0.70 was then applied to the held-out test set. The test set was reserved for final evaluation rather than threshold selection.

3.6 Multi-Seed and Control Experiments

For the study of sensitivity to data splits, multi-seed experiments used seed numbers 42, 21, and 7 on the same data. Sensitivity to split seeds was studied here, making it different from external evaluations independent of each other. The metrics of multi-seed classification such as F1 score were computed for a decision threshold of 0.5, which is different from 0.70 used in final experiments. A control experiment for

the hybrid model with aligned embeddings against hybrid model with shuffled embeddings was conducted to study the effect of alignment of text embeddings to the record embeddings in hybrid models. The TRANSFER-only rule was set up as a baseline by predicting the positive class for records with payment Type being `TRANSFER`. However, its PR-AUC cannot be compared with PR-AUC obtained from continuous model scores.

4. Experimental Results

4.1 Model Comparison

In the single seed-42 validation comparison, structured and hybrid XGBoost had similar PR-AUC, while the embedding-only and Isolation Forest baselines were near the positive-class prevalence. Across seeds, results did not show a reliable difference between structured and hybrid models.

Model	PR-AUC	ROC-AUC
Structured XGBoost	0.0837	0.8740
Hybrid XGBoost	0.0817	0.8703
Structured without `Type`	0.0226	0.5211
Hybrid without `Type`	0.0219	0.4971
Embeddings-only XGBoost	0.0212	0.4875
Isolation Forest	0.0210	0.4800

Table 1. Standardized validation-set model comparison.

4.2 Multi-Seed and Control Results

The following results report mean ± standard deviation across split seeds 42, 21, and 7. Only the split seed varies, and all runs use the same dataset. F1 is reported at the fixed threshold of 0.5. The TRANSFER-only rule's PR-AUC is based on binary predictions and is not directly comparable with score-based model PR-AUC.

Model	PR-AUC (mean ± SD)	ROC-AUC (mean ± SD)	F1 (mean)
Structured XGBoost	0.0802 ± 0.0033	0.8681 ± 0.0056	0.151
Hybrid (aligned embeddings)	0.0826 ± 0.0002	0.8696 ± 0.0018	0.148
Hybrid (shuffled embeddings)	0.0806 ± 0.0033	0.8688 ± 0.0048	0.148
TRANSFER-only rule	0.0810 ± 0.0031	0.8684 ± 0.0031	0.150
Embeddings-only	0.0226 ± 0.0010	0.4936 ± 0.0134	0.041
Structured without `Type`	0.0227 ± 0.0009	0.5006 ± 0.0149	0.040
Hybrid without `Type`	0.0225 ± 0.0006	0.4965 ± 0.0152	0.041

Table 2. Multi-seed comparison across three split seeds.

No model reliably outperforms the TRANSFER-only rule in these experiments. The hybrid's small PR-AUC advantage (0.0826 versus 0.0802) is within seed variation, and the shuffled-embedding control does not support a reliable embedding benefit. The detailed multi-seed results are saved in `results/multiseed_model_comparison_summary.csv`; shuffled-embedding and Type-control results are in `results/shuffled_embedding_and_type_control_summary.csv`.

4.3 Payment Type and Rule Baseline

Payment type	Rows	Suspected-fraud records	Suspected-fraud rate
TRANSFER	49,883	4,062	8.14%
CASH	19,616	0	0.00%
DEBIT	69,295	0	0.00%
PAYMENT	41,725	0	0.00%

All 4,062 positive labels occur in the TRANSFER category. On the same held-out test split, a rule flagging every TRANSFER order achieved precision 0.0805, recall 1.0000, and F1 0.1489, versus F1 0.1404 for the structured XGBoost model at threshold 0.70. The rule's PR-AUC is calculated from binary predictions, so it is not directly comparable with score-based model PR-AUC.

4.4 Final Test Evaluation

The structured XGBoost model was evaluated on the held-out test set using the validation-selected threshold of 0.70.

Metric	Test result
PR-AUC (Average Precision)	0.0765
ROC-AUC	0.8620
Precision	7.85%
Recall	66.47%
F1-score	0.1404
Decision threshold	0.70

Table 3. Final test-set performance (held-out test split).

The confusion matrix contained 28,978 true negatives, 6,492 false positives, 279 false negatives, and 553 true positives. Although the model identified approximately 66.5% of positive records, its precision was only 7.85%, indicating that most flagged records were not labelled as suspected fraud. Validation F1 was nearly flat (0.145–0.149) across thresholds from 0.05 to 0.70, so the selected threshold of 0.70 should not be read as a meaningful tuning gain; at threshold 0.5 the same model reached F1 0.1488 on this test split.

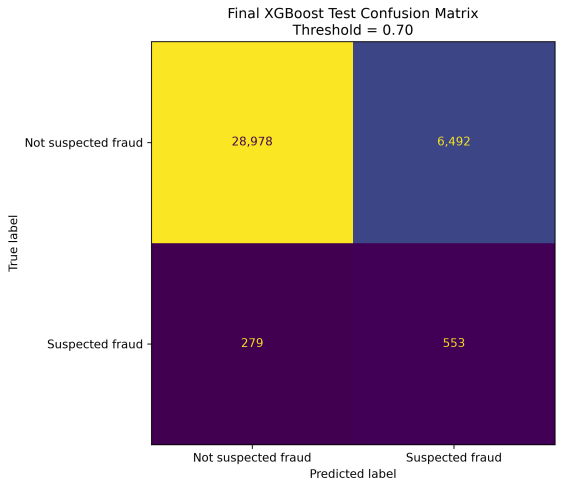


Figure 1. Test-set confusion matrix at threshold 0.70.

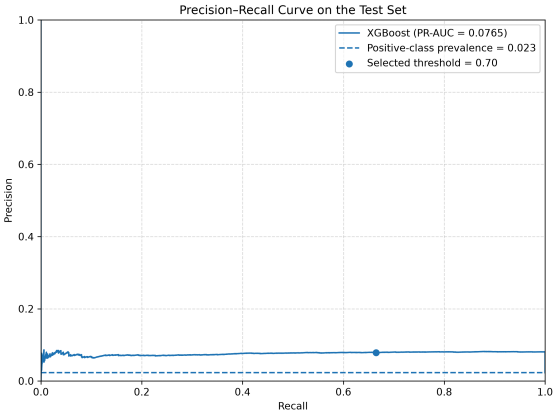


Figure 2. Test-set precision–recall curve; dashed line indicates positive-class prevalence.

5. Discussion

Multi-seed analysis of performance reveals no reliable difference between the structured version of XGBoost and the hybrid approach. The shuffled embedding control does not reveal any reliable advantage in using text embeddings either. All the positive labels have Type = TRANSFER, and the models that do not use this variable achieve near-chance performance levels. It would seem that the structure of the labels, rather than any additional semantics contained in the embeddings, drives the performance here. Because all eight embedded fields were also present as one-hot features in the structured model, the embeddings re-encoded information the baseline already had. These results therefore show that re-encoding short categorical fields adds no value here, and do not rule out gains from richer free text.

6. Limitations and Conclusion

The current study makes use of the SUSPECTED_FRAUD flag of the dataset instead of fraud outcomes which are independently verified. All 4,062 positive records have the Type TRANSFER, showing a strong link between the two for the dataset. Structured XGBoost, the MiniLM combination model, and TRANSFER flag alone perform equivalently with no demonstrated advantage to embeddings.

The current results are subject to limitations caused by the class imbalance, shorter text columns, and use of a single dataset. Further research must confirm the presence of features during prediction and consider richer textual data along with independent fraud verification.

References

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